Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

a. Team name: testofschool
b. Team leader name: Jung Min Kang
c. Problem Statement: People who pay per generation for AI video only learn a clip is unusable after they have paid for it.

Brief about the idea

Preflight: check an AI-video prompt before you pay

"you still burn credits on the failed attempt"

Trustpilot, Runway review, Sept 2026

"Only one of the generations was usable for my purposes"

Trustpilot, Sept 2026

"I'm tired of wasting valuable credits on bad footage, I simply can't use"

Adobe Community

"I recommend a lo-res preview render before having to spend actual credits"

Adobe Community

In a 2024 public benchmark (GenAI-Bench), 21% of 1,911 human pairwise votes rejected both videos.

Opportunities

Refunds cover generation errors, not finished clips that are unusable. Nothing sits before "pay".

Who pays today

Per-generation buyers. Runway TrustScore 1.1/5 from 340 reviews (Sept 2026). Adobe feature request: "I lose too many credits doing trial and error" (219 votes).

Who would buy a pre-pay step

Studios and agencies running generation pipelines. Generator platforms as an embedded step.

Built by a Seoul producer who ran generative-video pipelines for 2.5 years. Tested by a teammate in India.

Business model: free risk check; paid API per check for studios, agencies and generator platforms.

No revenue or market-size figure is claimed.

List of features offered by the solution

Preflight: three rungs before you pay

Risk check (free)

Failure-risk estimate from 1,911 human votes, with the nearest historical prompts as evidence.

Gemini rewrite

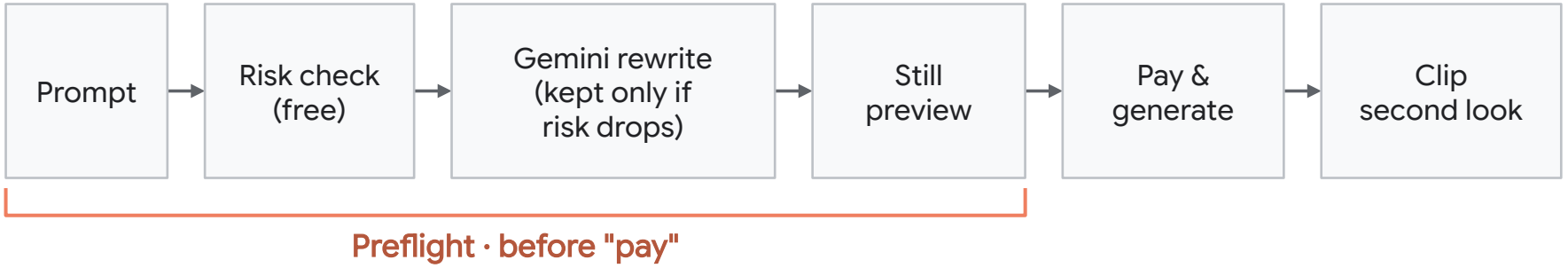
Kept only if the estimated risk drops.

A still before a $3.20 Veo attempt

Veo 3.1, 8 s at $0.40/s (Google pricing page).

After generation: upload a clip for a Gemini second look. Its verdict is stored separately as feedback.

Process flow diagram or Use-case diagram





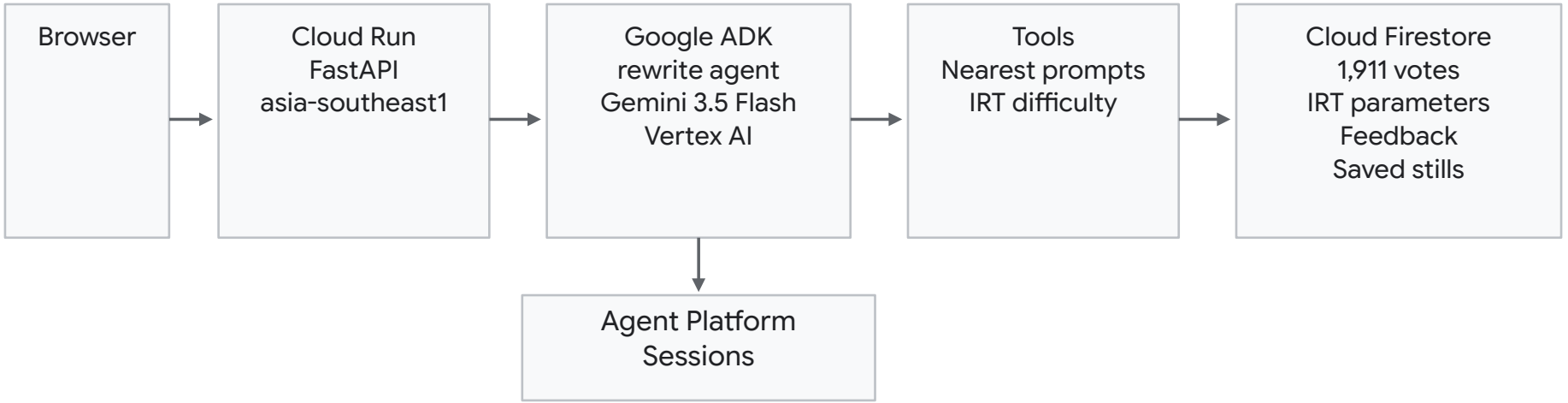
Saved original still



Saved rewrite still

Architecture diagram of the proposed solution

Browser request and Google Cloud services



Clip review: Gemini video understanding

Still preview: Gemini image model

Technologies to be used in the solution

Cloud Run (FastAPI, asia-southeast1)

Hosts the browser request workflow.

Google ADK + Gemini 3.5 Flash on Vertex AI

Rewrite agent with Agent Platform Sessions. Tools retrieve nearby prompts and IRT difficulty.

Cloud Firestore

Stores 1,911 votes, IRT parameters, feedback and saved stills.

Gemini video understanding + Gemini image model

Clip second look and still preview.

Active for sample prompts (saved stills); live generation limited

Snapshots of the prototype

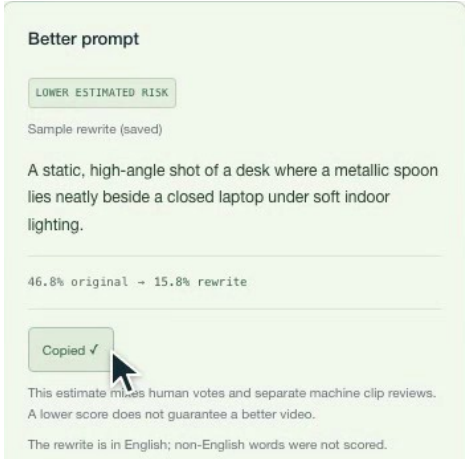
a spoon and a laptop: 46.8% → 15.8% estimated risk after the Gemini rewrite.

Risk + why



Text neighbours “a spoon and a laptop” and “a tv and a laptop” support an estimated risk score of 46.8%. The score uses nearby evidence and any separately stored clip feedback; it is not a calibrated probability that a current video model fails.

Better prompt



Saved still



0 model calls
$0.00

A still cannot judge motion, flicker or timing.

Prototype Performance report/Benchmarking

What we measured, and the limits

Held-out AUC 0.67

Historical signal, not a calibrated probability for today's models.

Live image generation

60 test calls across three Gemini image models. Best 8/20 under shared capacity.

Judges see saved stills for the 5 samples, labelled "0 model calls · $0.00".

Clip-review accuracy: not measured

Additional Details/Future Development (if any)

Next

Live stills with dedicated image capacity.

Separate tabs for check and clip review.

Non-English vote data before scoring non-English prompts.

On-page emphasis of the risk drop.

Provide links to your:

GitHub Public Repository

https://github.com/testofschool/preflight

Demo Video Link (3:28)

https://youtu.be/nO-J4-5G-Gs

Final Product Link

https://preflight-jcgro45veq-as.a.run.app